



INTERNSHIP PROJECT PRESENTATION

AI Admission Predictor

College Admission Prediction using Machine Learning

SUBMITTED BY

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B.Tech AI & Data Science, II Year

GUIDE

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ORGANISATION

KanmalaiTech UK Limited
ML Intern · Online (Remote)

PERIOD

3 – 28 August 2026 (4 weeks)
HOD: Dr. P. Nagarajan

Python

Numpy

pandas

scikit-learn

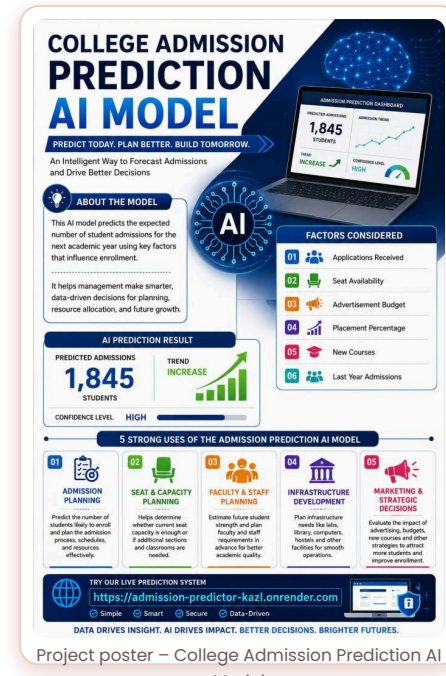
Flask

Chart.js

GitHub

Render

J.N.N Institute of Engineering · Dept. of Artificial Intelligence & Data Science



Project poster – College Admission Prediction AI Model

Internship Overview & Roadmap



J·N·N
INSTITUTE OF
ENGINEERING

4

Weeks (Online)

8

Modules

7

Input Factors

1

Live Web App

OBJECTIVES

Learn ML basics & Python (NumPy, pandas) • train & evaluate regression models • build, deploy & present a 7-factor admission predictor

4-WEEK WORKFLOW — FROM FUNDAMENTALS TO A LIVE PRODUCT

WEEK 1 · 3–7 Aug

Module 1

ML fundamentals, Python set-up,
NumPy

WEEK 2 · 10–14 Aug

Module 2

pandas DataFrames, 200-row
dataset

WEEK 3 · 17–21 Aug

Module 3

Train/test split, Linear Reg. vs Random
Forest

WEEK 4 · 24–28 Aug

Modules 4 – 8

Web app, prediction service, charts,
deployment, poster

THE 8 MODULES

M1 ML Fundamentals & Python

M5 Prediction Service (Flask)

M2 Data Handling (pandas)

M6 Result Visualisation

M3 Model Training & Evaluation

M7 Deployment & GitHub

M4 Input Interface (web form)

M8 Project Poster

PROBLEM STATEMENT

Colleges estimate admissions from last year's figure or management opinion, ignoring factors that act together — leading to **empty seats** or **overloaded facilities**. **Goal:** turn a handful of admission factors into a quick, easy-to-read ML prediction for planning seats, staff and infrastructure.



HOW THE AI ADMISSION PREDICTOR WORKS



7 INPUT FACTORS (TYPICAL RANGE)

Applications Received	1,000 – 3,000
Seats Available	800 – 2,200
Placements (%)	55 – 98
Advertisement Budget	₹80,000 – ₹8,00,000
Courses Available	5 – 12
Annual Fees	₹75,000 – ₹2,50,000
Last Year Admissions	similar to seats range

MANUAL ESTIMATION VS ML PREDICTOR

Aspect	Manual	AI Admission Predictor
Factors	1–2 (last year)	7 factors together
Consistency	Varies by person	Same input, same answer
Speed	Meetings & calculation	Result in a moment
Explanation	Opinion-based	Charts for clarity
Updating	Redone by hand	Retrain with new data
Access	Depends on people	Online, any time

RESPONSIBLE USE

Only institution-level figures are used — no personal applicant details. The output is **decision support for planning**, never for selecting or rejecting individual applicants.



ML PIPELINE FOLLOWED IN THE INTERNSHIP



TRAIN-TEST SPLIT (PRACTICE DATASET)

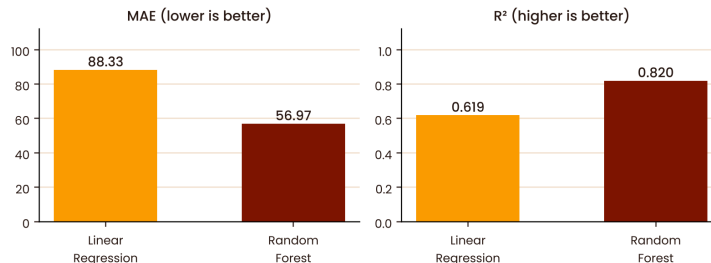
160 train

40 test

200 rows × 8 columns · no missing values

- ▶ **Linear Regression** – straight-line, easy to interpret
- ▶ **Random Forest** – many trees averaged; captures interacting factors
- ▶ Testing on unseen rows guards against **overfitting**
- ▶ **MAE**: lower is better · **R²**: closer to 1 is better

MODEL COMPARISON ON THE 40 TEST ROWS



Random Forest has the smaller error and explains more variation, so it powers the predictor. Figures are from the generated practice dataset – not the accuracy of the deployed app.

Linear Regression

MAE 88.33 · R² 0.619

Random Forest ✓

MAE **56.97** · R² **0.820**

TOOLS & TECHNOLOGIES

Python 3

NumPy

pandas

scikit-learn

Flask API

HTML/CSS/JS UI

Chart.js

Git/GitHub

Render



AI Admission Predictor

Admission Predictor

Applications Received
1200
Typical range: 1000-2000

Seats Available
900
Typical range: 800-2200

Placements (%)
86
Typical range: 75-95

Advertisement Budget (₹)
40000
Typical range: 30000-50000

Courses Available
8
Typical range: 5-12

Annual Fees (₹)
19995
Typical range: ₹1,00,000-₹2,00,000

Last Year Admissions
800
Previous typical seats range

Predict Admissions

Input form – 7 fields with typical ranges & Predict button

731

Predicted

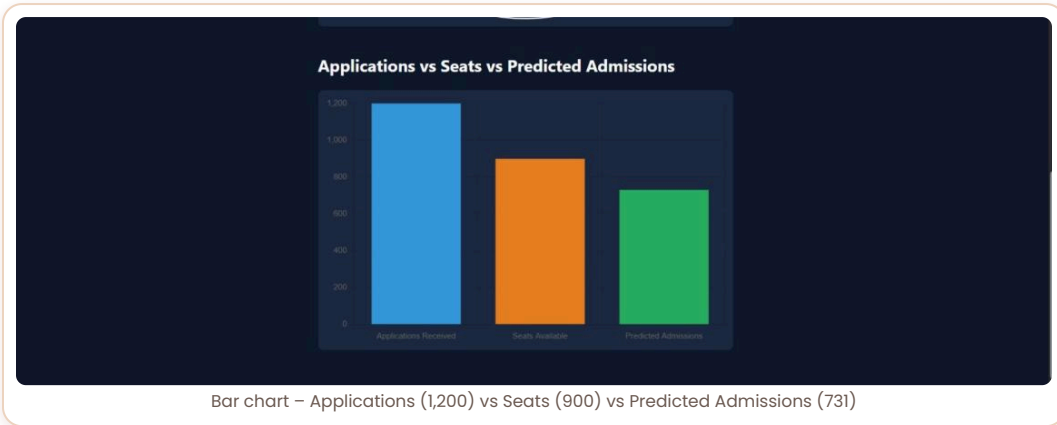
81.2%

Seat utilisation

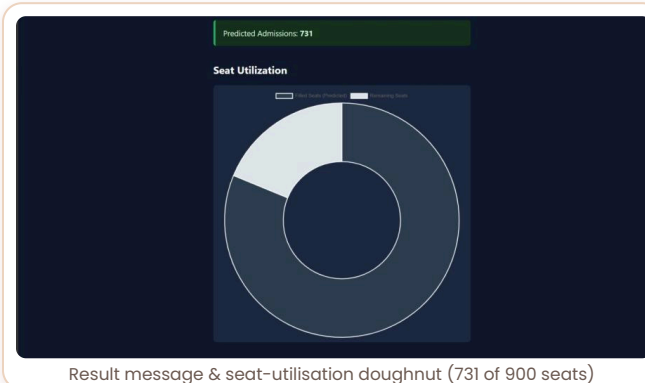
169

Seats remaining

Prediction (731) is lower than last year's 800 – not just a copy of the previous figure.



Bar chart – Applications (1,200) vs Seats (900) vs Predicted Admissions (731)



Result message & seat-utilisation doughnut (731 of 900 seats)

SAMPLE INPUT USED

Applications	1,200
Seats	900
Placements	86%
Ad budget	₹4,00,000
Courses	8
Annual fees	₹1,99,985
Last year	800



DEPLOYMENT FLOW — FROM CODE TO LIVE APPLICATION



SKILLS GAINED

- ▶ **ML:** supervised learning, regression, MAE & R^2
- ▶ **Python:** NumPy, pandas, scikit-learn
- ▶ **Web:** form interface & Flask-style prediction service
- ▶ **Visualisation:** bar & doughnut charts
- ▶ **Deployment:** Git, GitHub, Render
- ▶ **Communication:** poster & technical report

CONCLUSION

- ▶ Complete hands-on journey through the ML workflow in 4 weeks
- ▶ 7 admission factors → predicted admissions with clear charts
- ▶ Deployed online, code on GitHub, poster prepared
- ▶ Gives useful **decision support** when data is handled carefully

LIMITATIONS

Practice data is generated; only 7 factors; no database.

FUTURE WORK

- ▶ Train on real multi-year records from colleges
- ▶ Show MAE / R^2 inside the app
- ▶ Database & login to save and compare years
- ▶ More factors: cut-off marks, location, nearby competitors
- ▶ Automatic yearly retraining

Thank You

Live project: admission-predictor-kazl.onrender.com

J. Sam Jacob · Reg No 110725102213 · Certificate of Successful Completion issued 01-Sep-2026